

Fall 2014

Problem 1 (Fa14). Let $a(n)$ be the number of ways that Harry Potter can buy a new broomstick valued at n knuts using bronze knuts, silver sickles worth 29 knuts, and gold galleons worth 17 sickles. Find $\sum_n a(n)z^n$ and $\lim_{n \rightarrow \infty} a(n)/n^2$.

Problem 2 (Fa14). Suppose that f_n is a sequence of nonnegative continuous functions on the unit interval. Find counterexamples to three of the following four inequalities:

$$\limsup_{n \rightarrow \infty} \int_0^1 f_n(x) dx \leq \int_0^1 \limsup_{n \rightarrow \infty} f_n(x) dx$$

$$\limsup_{n \rightarrow \infty} \int_0^1 f_n(x) dx \geq \int_0^1 \limsup_{n \rightarrow \infty} f_n(x) dx$$

$$\liminf_{n \rightarrow \infty} \int_0^1 f_n(x) dx \leq \int_0^1 \liminf_{n \rightarrow \infty} f_n(x) dx$$

$$\liminf_{n \rightarrow \infty} \int_0^1 f_n(x) dx \geq \int_0^1 \liminf_{n \rightarrow \infty} f_n(x) dx$$

Problem 3 (Fa14). Suppose that f is a twice differentiable real valued function on the real line such that $|f(x)| \leq 1$ and $|f''(x)| \leq 1$ for all x . Find a constant b such that $|f'(x)| < b$ for all x .

Problem 4 (Fa14). Let D be the set consisting of the open unit disc together with the point 1. Show that the power series

$$\sum_{n>0} \frac{z^{3^n}}{n} - \frac{z^{2 \cdot 3^n}}{n}$$

converges at all points of D . By examining points with argument of the form $\pi/3^k$, show that the function it converges to is not continuous.

Problem 5 (Fa14, Fa15, Fa21). 1. Suppose $p(z) = c(z - a_1) \cdots (z - a_n)$ is a complex polynomial. If z has positive real part and all the zeros a_i have negative real part, show that $p'(z)/p(z)$ has positive real part.

2. Show that all the zeros of the derivative p' of a complex polynomial lie in the convex hull of the zeros of p .

Problem 6 (Fa14). Find the order of the group of linear transformations preserving a nondegenerate skew-symmetric bilinear form on a 4-dimensional vector space over the field with 3 elements, and find the number of such forms.

Problem 7 (Fa14). Find a basis of the intersection of the subspace of $\mathbb{R}^4$ spanned by the vectors $\{(1, 1, 0, 0), (0, 1, 1, 0), (0, 0, 1, 1)\}$ and the subspace spanned by the vectors $\{(1, 0, t, 0), (0, 1, 0, t)\}$, where t is given.

Problem 8 (Fa14). Let X be a totally ordered set, i.e., equipped with a non-reflexive, transitive binary relation $<$ such that for every $x \neq y$, either $x < y$ or $y < x$. Let $L(X)$ denote the set of subsets $S \subseteq X$ with the property that for all $y \in S$ and $x < y$, $x \in S$. Find:

1. a countably infinite totally ordered set X for which $L(X)$ has the smallest possible cardinality;
2. a countably infinite totally ordered set X for which $L(X)$ has the largest possible cardinality.

Problem 9 (Fa14, Sp25). 1. By counting the number of pairs (g, x) with $g \in G$, $x \in X$ and $g(x) = x$, show that the number of orbits of a finite group G acting on a finite set X is the average number of fixed points of elements of the group.

2. In how many ways, up to symmetries of the hexagon, can one color the vertices of a regular hexagon using 4 colors?
3. In how many ways, up to symmetries of the square, can one color the 4 corners of a square with 10 colors?

Problem 10 (Fa14). Show that the polynomial

$$p_n(x) = 1 + \frac{x^1}{1!} + \cdots + \frac{x^n}{n!}$$

has exactly one real zero if n is odd and none if n is even.

Problem 11 (Fa14). Either prove or describe a counterexample to the following statement: “If a continuous real valued function on the plane is bounded on all straight lines then it is bounded”.

Problem 12 (Fa14). Let $f : [0, 1] \times [0, 1] \rightarrow \mathbb{R}$ be continuous and assume that for all $x \in [0, 1]$ there is a unique y_x such that $f(x, y_x) = \max\{f(x, y) \mid y \in [0, 1]\}$. Let $g(x) = y_x$. Show that $g : [0, 1] \rightarrow [0, 1]$ is continuous.

Problem 13 (Fa14). Find four power series f_1, f_2, f_3, f_4 with radius of convergence 1 such that f_1, f_2 converge at 1 but f_3, f_4 do not, and the functions given by f_1, f_3 can be extended to functions holomorphic in a neighborhood of 1, but the functions given by f_2, f_4 cannot be.

Problem 14 (Fa14). Let f be a doubly periodic meromorphic function: $f(z+1) = f(z) = f(z+i)$ for all $z \in \mathbb{C}$. Let z_a be the zeros of f inside the unit square $0 < \Re z, \Im z < 1$, w_b be its poles inside the square, and k_a and l_b be respective multiplicities. Assuming that f has no zeros or poles on the boundary of the square, prove that

$$\sum_a k_a z_a - \sum_b l_b w_b \in \mathbb{Z}[i]$$

that is, is a Gaussian integer.

Problem 15 (Fa14). Show that there is a sequence of 4×4 matrices $A(n)$ with real entries, which converges to

$$A = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

and such that $A(n)$ has 4 distinct real eigenvalues, two of which are positive and two negative.

Problem 16 (Fa14). Let n be a fixed positive integer, and define two $n \times n$ real symmetric matrices A and B to be equivalent if there is a nonsingular real matrix C with $CAC^T = B$. How many equivalence classes are there?

Problem 17 (Fa14). Determine, up to isomorphism, all finite groups G such that G has exactly three conjugacy classes.

Problem 18 (Fa14, Fa24). How many zeros does the polynomial $x^{100000} - 1$ have in the finite field $\mathbb{F}_{65537}$? Observe that $65537 = 2^{16} + 1$ is a prime.